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INTERNSHIP REPORT ON

"DESIGN AND CAD MODELLING OF LANDING GEAR CONFIGURATION FOR A TYPICAL AIRCRAFT"

by

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at

CSIR- NATIONAL AEROSPACE LABORATORIES
BANGALORE, 560017

CSIR-NAL

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BACHELOR OF TECHNOLOGY
in
AEROSPACE ENGINEERING

CERTIFICATE

This is to certify that SALMAN FARIS E T, USN No. 22BTRAS038 student of **JAIN UNIVERSITY** pursuing **B Tech Aerospace** has successfully completed Internship titled "**DESIGN AND CAD MODELLING OF LANDING GEAR CONFIGURATION FOR A TYPICAL AIRCRAFT**" during **16-02-2026 to 16-05-2026** under the guidance of SUDHAKAR, S. Shri**, Principal Scientist, ACD** towards the fulfilment of the requirements of their university.

We wish SALMAN FARIS E T the very best in all his future endeavors.

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Internship Guide
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ABSTARCT

I spent my internship at CSIR-National Aerospace Laboratories working on the landing gear for a High-Altitude Pseudo Satellite (HAPS) aircraft. The plane's got a tricycle arrangement — two main gears under the wings, and a rear gear near the back of the fuselage.

The goal was pretty clear: create a landing gear system that's light, tough, stable, easy to build, and good at absorbing the impact when the aircraft lands. To make this happen, I designed three different main landing gear setups, built them out in CAD, and compared how they held up. Along the way, I came up with a rear gear to steady the tail while the plane is on the ground or coming in for landing.

Every main gear design used its own suspension system to soften the landing. I switched up the structural layouts and linkages to see how each dealt with forces, stood up to stress, and whether you could actually build it without too much hassle.

After running through all the comparisons, Configuration 1 came out ahead. It had the best structural strength, handled loads better, gave the aircraft more stability, and soaked up shocks more efficiently than the other designs. That's why it became the final pick.

Everything got modeled in CATIA Part Design Workbench. Most of my effort went into shaping the basic geometry, figuring out the structure, slotting in wheels and suspension, and checking if each setup made sense from a practical angle.

This report walks you through the fundamentals of landing gear, the specifics of the tricycle layout, how I tackled the CAD work, details on each configuration, and how I landed on the most suitable design for the aircraft.

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1. OBJECTIVE

- Study what a HAPS aircraft needs when it comes to tricycle landing gear.
- Come up with three different main landing gear setups, each with suspension systems that help handle impact loads.
- Design a lightweight rear gear for the back of the aircraft.
- Build precise 3D CAD models of these landing gear assemblies in CATIA Part Design Workbench.
- Compare each design in terms of structural stability, how it spreads loads, and whether it fits the operational demands.
- Pick the landing gear configuration that works best for the HAPS aircraft.
- Get hands-on experience in aerospace CAD modeling and structural design for landing gear.

2. INTRODUCTION

2.1 Overview of HAPS Aircraft

High Altitude Pseudo Satellites, or HAPS, are these super light, long-lasting planes that can fly way up in the atmosphere, hanging out there for ages. They're basically floating platforms in the sky, handling everything from relaying communications and running surveillance to keeping an eye on the environment, collecting atmospheric data, or helping out with strategic aerospace stuff.

HAPS aren't just any aircraft. Their design zeroes in on a few key points: they have to be as light as possible, glide efficiently, stay in the air for a long time, sip energy instead of guzzling it, and handle slow speeds without getting wobbly.

Because HAPS are so lightweight, you can't just slap on any component and call it a day. Every part especially the structural systems needs to be trimmed down and fine-tuned so the plane stays light but still does its job reliably. The landing gear, for instance, isn't just an afterthought. It's a big deal, since it handles all the action when the plane is on the ground taxiing, taking off, and landing.

During the internship, I worked on a HAPS aircraft that used a tricycle landing gear setup: two main gears under the wings, and one in the back near the tail. The landing gear had to tick a bunch of boxes: good ground clearance, solid stability, gentle landings, and dependable load distribution whenever the plane touched or moved along the ground.

2.2 Importance of Landing Gear Systems

The landing gear system is absolutely crucial for any aircraft. It keeps the plane steady during ground operations and takes on the heavy loads when landing. Basically, it has to do a lot: hold up the aircraft's weight, absorb the shock of landing, keep things stable while taxiing, and make sure takeoffs and landings are smooth. It also needs to protect the structural integrity through countless cycles and provide enough ground clearance so the aerodynamic surfaces don't scrape.

When you are dealing with lightweight aerospace vehicles like HAPS aircraft the landing gear faces an extra tough challenge. You have to find the sweet spot between keeping the mass low and making sure it stays rigid and strong. The gear also has to absorb energy well, have a simple design, be easy to manufacture and assemble, and need minimal maintenance.

Getting this right means paying close attention to the details: geometry, how everything links together, where you put the wheels, and how the suspension fits in. If you don't think about each bit, the whole system could fall apart literally.

3. TRICYCLE LANDING GEAR CONFIGURATION

3.1 General Arrangement

The HAPS aircraft used a tricycle landing gear setup, which means it had two main landing gears under the wings and a smaller one near the tail. You see this design in most modern planes because it's more stable, spreads the weight better, and keeps landings safer.

With the tricycle layout, pilots get easier control on the ground, steadier handling, and a lower chance of smacking the tail on landing. Braking works better, landing feels safer, and it's just simpler to manage the plane overall.

Most of the aircraft's weight sits on the two main gears, while the rear gear supports the tail and keeps everything balanced..

3.2 Main Landing Gear Requirements

The main landing gear assemblies were required to satisfy multiple structural and operational requirements.

The major requirements included:

- Supporting the majority of aircraft weight
- Absorbing landing impact loads
- Maintaining lightweight structural configuration
- Providing sufficient rigidity and strength
- Ensuring compatibility with aircraft geometry
- Supporting smooth taxiing operations
- Reducing stress concentration during touchdown

To satisfy these requirements, three separate landing gear concepts were developed and comparatively evaluated. All three configurations incorporated suspension systems intended to absorb impact loads and improve landing performance.

3.3 Rear Landing Gear Requirements

The rear landing gear was designed primarily to:

- Support the tail section during ground operations
- Maintain aircraft balance
- Prevent structural damage to aft fuselage components
- Improve overall ground stability

Unlike the main landing gear assemblies, the rear landing gear did not incorporate a suspension mechanism because the expected load conditions were comparatively lower.

The rear landing gear design emphasized:

- Structural simplicity
- Lightweight geometry
- Ease of manufacturing
- Reduced component count

4. CAD MODEL DEVELOPMENT

4.1 Design Methodology

We started by sketching out some ideas for the landing gear, and then moved on to figuring out how everything would fit together structurally. Next, we worked out the geometry, turning those concepts into layouts that actually made sense. After that, we built CAD models to see if the designs held up both structurally and visually making sure they'd really work for a HAPS aircraft. Once we had those models, we compared the options, weighed the pros and cons, and picked the best configuration.

The CAD phase was all about creating concepts that made sense from both an engineering and integration standpoint, not getting bogged down in the nitty-gritty details of manufacturing. The main focus stayed on developing strong, clear structural layouts rather than perfecting every small part for production.

4.2 CAD Software Usage

We used CATIA for all our CAD modeling work, sticking to the Part Design Workbench. The development process covered a lot: creating structural members, placing joints, setting up suspension linkages, positioning the wheel assemblies, checking ground clearance, and fine-tuning the geometry. CATIA V5 made it easy to build precise models and visualize the landing gear concepts, and we kept everything in line with solid aerospace design standards..

5. MAIN LANDING GEAR CONFIGURATIONS

5.1 Configuration 1 : Main Landing Gear Concept 1

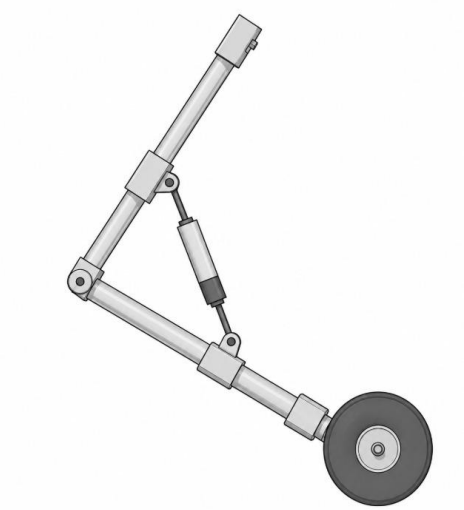


Figure 1 : Main landing gear configuration 1

5.1.1 Configuration Description

The first landing gear configuration incorporated a suspension mechanism connected between the upper structural member and the lower wheel-supporting member.

The configuration consist of :

- Primary inclined support member
- Secondary lower support member
- Pivot joints
- Wheel assembly
- Suspension/shock absorber element
- Connecting linkage arrangement

The suspension system was integrated to improve impact absorption capability and reduce the transmission of landing loads to the aircraft structure.

5.1.2 Structural Arrangement

This setup used a triangular linkage, and that made everything sturdier. It handled loads better, controlled movement during touchdown, and cut down on stress in the structure.
They placed the suspension right between the top and bottom parts so it could absorb impact energy as efficiently as possible.

Thanks to the way the linkage worked, the structure had more flexibility during landings. Instead of dumping all the force onto one component, it spread the load across several parts, so nothing got overloaded.

5.1.3 Advantages of Configuration 1

- Shock Absorption Capability : The integrated suspension system significantly reduced landing impact loads transferred to the airframe.
- Improved Structural Flexibility : The linkage mechanism allowed controlled deformation and smoother landing response.
- Better Load Distribution : Loads were distributed through multiple interconnected structural members.
- Enhanced Landing Performance : The suspended arrangement improved touchdown stability and operational safety.
- Reduced Structural Stress : The configuration minimized localized stress concentrations during impact loading.

5.1.4 Limitations

The setup brought some clear benefits, but it wasn't without its headaches. There's more structural complexity now, extra moving joints to think about, and the system weighs a bit more. Plus, getting the alignment of all the linkage parts right takes some real precision. Still, the upsides easily outweigh these drawbacks.

5.2 Configuration 2 – Simplified Main Landing Gear with Suspension

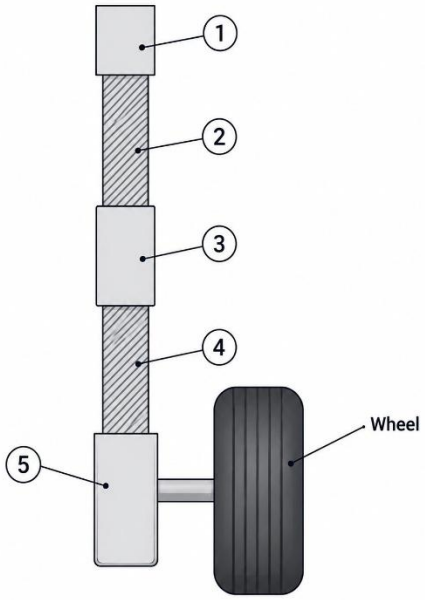


Figure 2 : Main landing gear configuration 2

5.2.1 Configuration Description

The second landing gear configuration utilized a simplified straight-member arrangement integrated with a suspension system.

The configuration mainly included:

- Upper mounting section
- Straight structural members
- Intermediate support blocks
- Wheel mounting region
- Internal suspension arrangement

This concept emphasized simplicity and lightweight design.

5.2.2 Design Characteristics

The simplified geometry reduced the number of components and provided:

- Easier manufacturing
- Reduced assembly complexity
- Lower maintenance requirements
- Reduced structural weight

The straight-member arrangement also simplified the overall structural layout.

5.2.3 Advantages

The major advantages included:

- Lightweight structural arrangement
- Simpler manufacturing process
- Lower fabrication complexity
- Easier integration with aircraft structure
- Reduced mechanical complexity

5.2.4 Limitations

Although this configuration incorporated a suspension mechanism, the simplified geometry resulted in :

- Lower structural flexibility compared to Configuration 1
- Reduced load distribution efficiency
- Higher localized load transfer during touchdown
- Reduced impact dissipation capability

Due to these limitations, Configuration 2 was not selected as the final preferred arrangement.

5.3 Configuration 3 – Fork-Type Main Landing Gear with Suspension

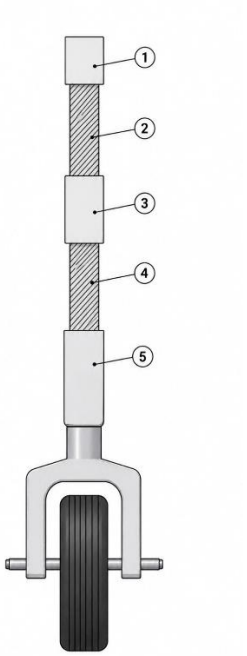


Figure 3 : Main landing gear configuration 3

5.3.1 Configuration Description

The third configuration utilized a fork-type wheel support arrangement connected to a vertically aligned structural member.

The configuration included:

- Upper attachment section
- Main vertical strut
- Fork-type wheel support
- Suspension arrangement
- Wheel axle assembly

This arrangement provided improved wheel support symmetry and better wheel alignment characteristics.

5.3.2 Structural Arrangement

The fork-type geometry enabled:

- Improved wheel positioning
- Better lateral stability
- Symmetric load transfer

- Improved wheel support rigidity

The suspension mechanism was integrated into the primary strut arrangement.

5.3.3 Advantages

Major advantages of the configuration included:

- Improved wheel support stability
- Better structural symmetry
- Enhanced wheel alignment
- Improved geometric balance

5.3.4 Limitations

The configuration also introduced:

- Increased structural weight
- Higher manufacturing complexity
- Larger component count
- Increased assembly requirements

Although structurally stable, the configuration was comparatively more complex than the selected design.

6. REAR LANDING GEAR CONFIGURATION



Figure 4 : Rear landing gear

6.1 Rear Landing Gear Description

The rear landing gear was designed to support the aft section of the aircraft during ground operations.

The design consisted of :

- Vertical support member
- Wheel support fork
- Wheel assembly
- Upper mounting interface

Unlike the main landing gear concepts, the rear landing gear did not incorporate a suspension mechanism.

6.2 Design Characteristics

The rear landing gear emphasized:

- Lightweight geometry
- Structural simplicity

- Reduced component count
- Ease of manufacturing

Since the rear gear carried comparatively lower operational loads, a rigid arrangement was considered sufficient.

6.3 Advantages

The rear landing gear offered:

- Simple structural design
- Lower manufacturing complexity
- Reduced overall system weight
- Easier maintenance and integration

7. COMPARATIVE EVALUATION OF MAIN LANDING GEAR CONFIGURATIONS

We compared the three main landing gear concepts, looking at things like structural stability, how well they distribute loads, how suspension fits in, structural efficiency, simplicity of their shapes, how practical they are for operation, and how easy they are to manufacture.

Here's what stood out: Configuration 1 handled load transfer best and performed really well during landing. Configuration 2 was the simplest of the bunch but didn't adapt as well to different situations. Configuration 3 gave stronger support to the wheels, but the design was a lot more complicated.

All things considered, Configuration 1 came out on top. That's the one we picked for the final landing gear design.

8. SKILLS ACQUIRED

1. Software: During the internship, advanced surface modeling using CATIA was learned to generate doubly curved aircraft parts. DraftSight was utilized for 2D drawing creation.
2. Aerospace drawing standard: An aerospace drawing standard is a set of guidelines and specifications regarding the format and conventions for creating technical drawings in the aerospace industry. These standards ensure design intent, manufacturing specifications, and assembly instructions for aircraft components and systems.

9. CHALLENGES AND SOLUTIONS

During the internship, one of the major challenges faced was the absence of detailed 2D reference sketches for the landing gear models. Initially, it was difficult to estimate suitable dimensions, structural proportions, and component arrangements while creating the CAD models in CATIA. This issue was resolved by discussing the design requirements with experienced employees and mentors, which helped in understanding approximate dimensions and improving the overall structural layout and modeling accuracy.

10. CONCLUSION

The internship work successfully achieved the conceptual design and CAD modeling of landing gear systems for a HAPS aircraft at CSIR-National Aerospace Laboratories.

Three different main landing gear configurations incorporating suspension systems were developed and comparatively evaluated. Additionally, a lightweight rear landing gear without suspension was designed for aft fuselage support.

The complete modeling work was performed using CATIA Part Design Workbench. The project provided significant exposure to aerospace CAD modeling methodologies, structural arrangement development, and conceptual aircraft subsystem design.

Among the developed concepts, Configuration 1 demonstrated the best combination of structural stability, shock absorption capability, operational suitability, and load distribution performance. Therefore, it was selected as the preferred landing gear configuration for the HAPS aircraft.

The internship significantly improved understanding of landing gear systems, aerospace structures, CAD modeling workflows, and practical engineering design methodologies.